

Stock-path simulation · bar charts

Friday 17 March 2023 monthly expiry · 1-hour lookback · SPY / AAPL / MSFT

What this PDF is for

- This PDF is the bar-chart companion to the stock-path simulation tables for 17 March 2023.
- Each page is a 2×2 of the four study cells: 1-day vs 5-day, and hourly vs minutely rolling.
- Four bars per panel: GBM, Merton, GARCH, GARCH-Merton.
- Heston and Heston-Merton are excluded from every chart. Their errors are larger on this 1-hour / 1-minute setup and would squash the other bars; the values remain in the tables PDF.
- The outlined bar is the best model on that panel (lowest RMSE or MAE; highest ICP).

Methodology

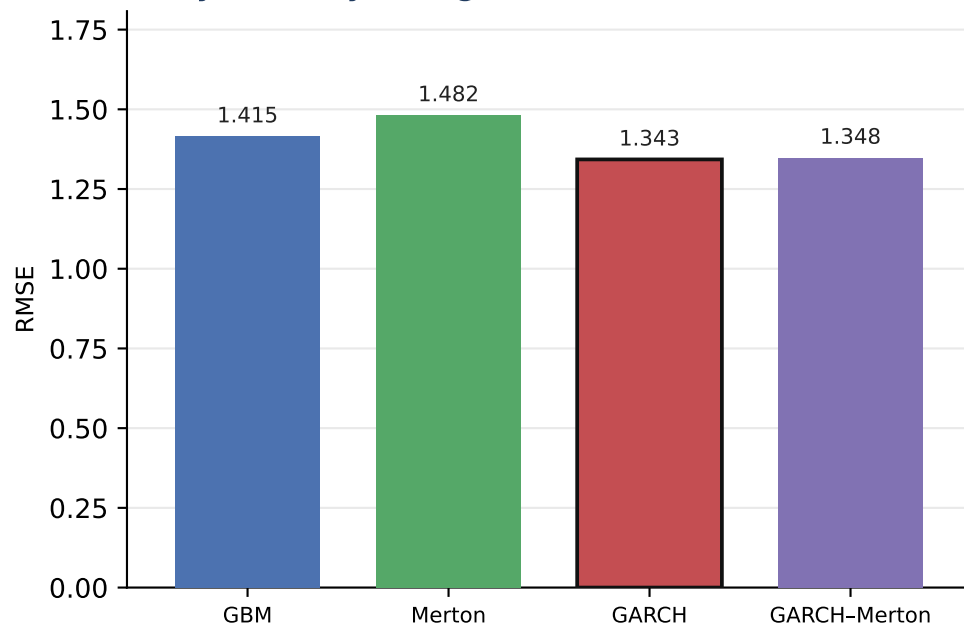
- Lookback is 1 hour (60 consecutive regular-trading-hours minutes) at every calibration time.
- Rolling is either hourly (parameters held for one hour) or minutely (parameters updated every minute).
- Models: GBM, Merton jump-diffusion, Heston (no jumps), Heston-Merton, GARCH(1,1), and GARCH-Merton. Same 1-minute notebooks as the rest of V2.
- Each name is simulated with 1,000 Monte Carlo paths and a common seed of 42.
- The reported path is the 50th percentile across paths at each minute — not the Monte Carlo mean.
- RMSE and MAE compare that median path with historical S_t on the continuous RTH clock (overnight and weekend gaps removed).
- ICP is the share of historical prices that fall inside the 25th–75th percentile band.
- Average band width is the mean of (75th – 25th percentile) over the window.

Heston and Heston-Merton are in the tables PDF only.

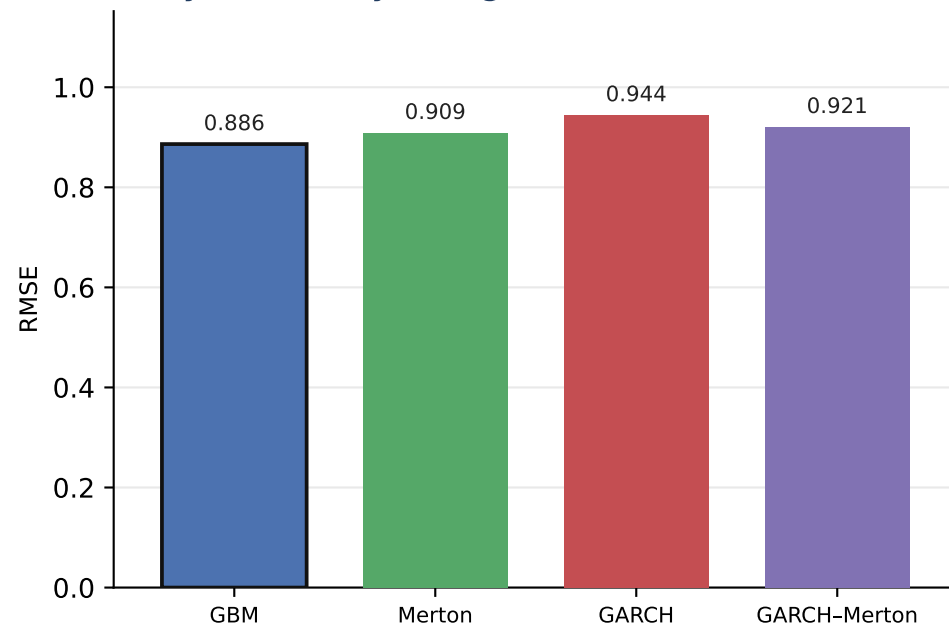
Windows: 1-day = Fri 17 Mar 2023 09:30–15:59 ET (RTH). 5-day = Mon 13 Mar – Fri 17 Mar 2023 (five RTH sessions).

RMSE of median path vs S_t · mean of SPY / AAPL / MSFT

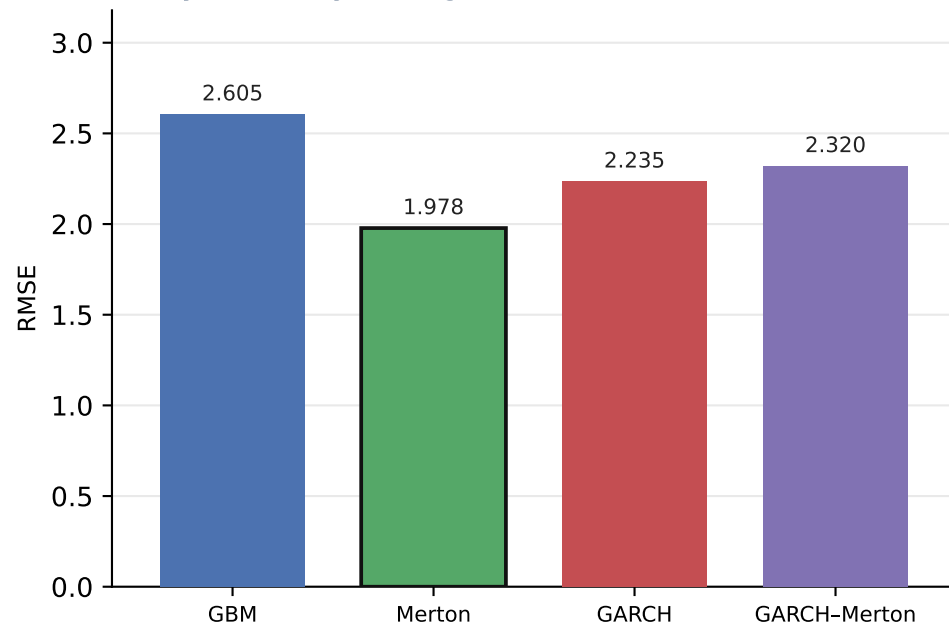
1-day · hourly rolling



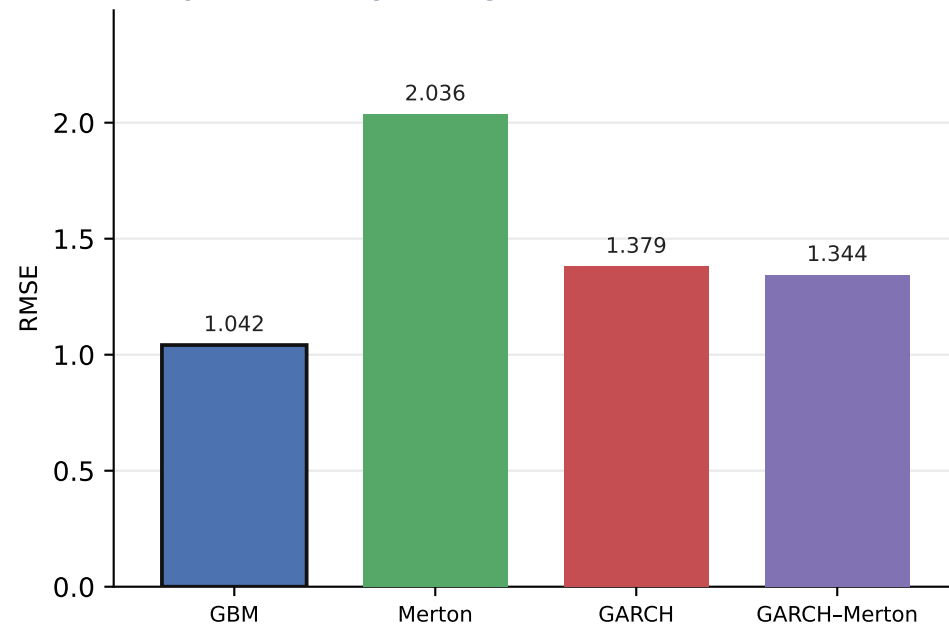
1-day · minutely rolling



5-day · hourly rolling



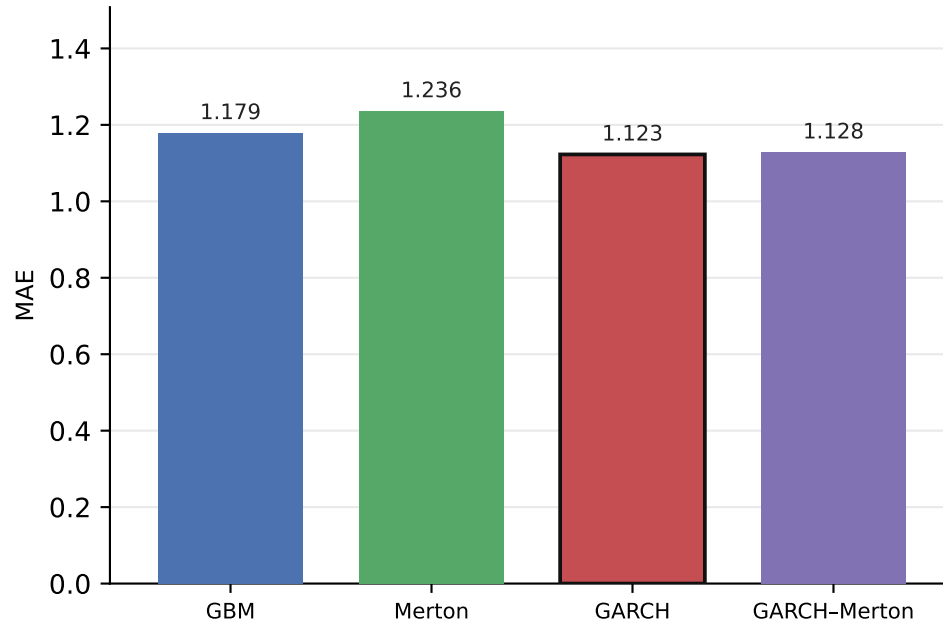
5-day · minutely rolling



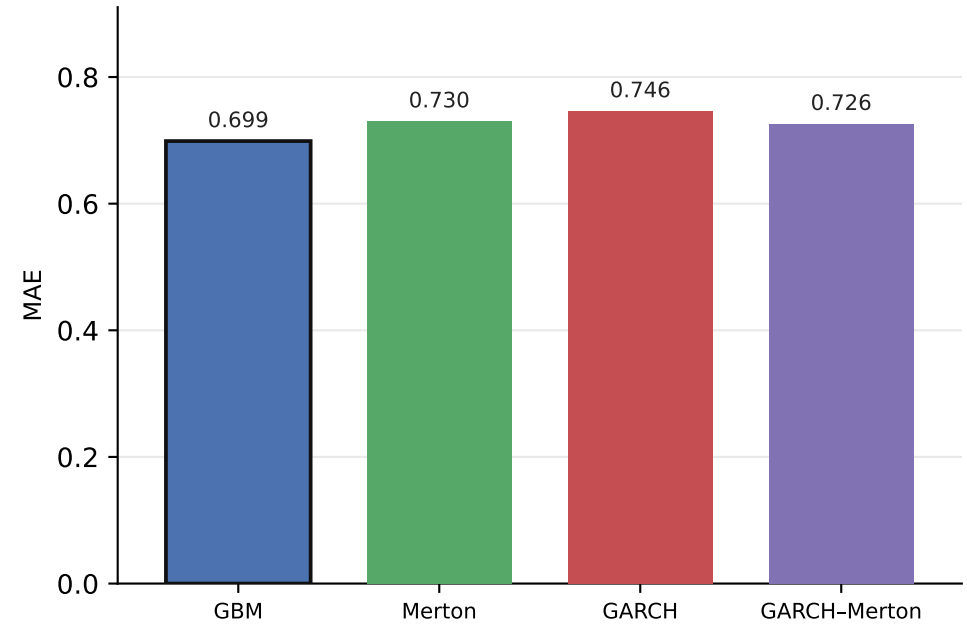
Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

MAE of median path vs S_t · mean of SPY / AAPL / MSFT

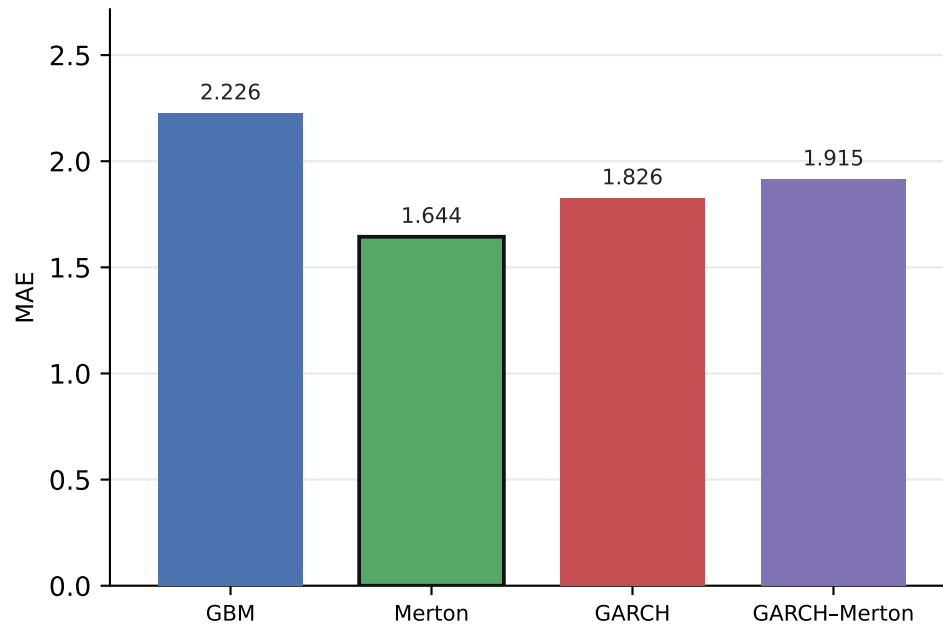
1-day · hourly rolling



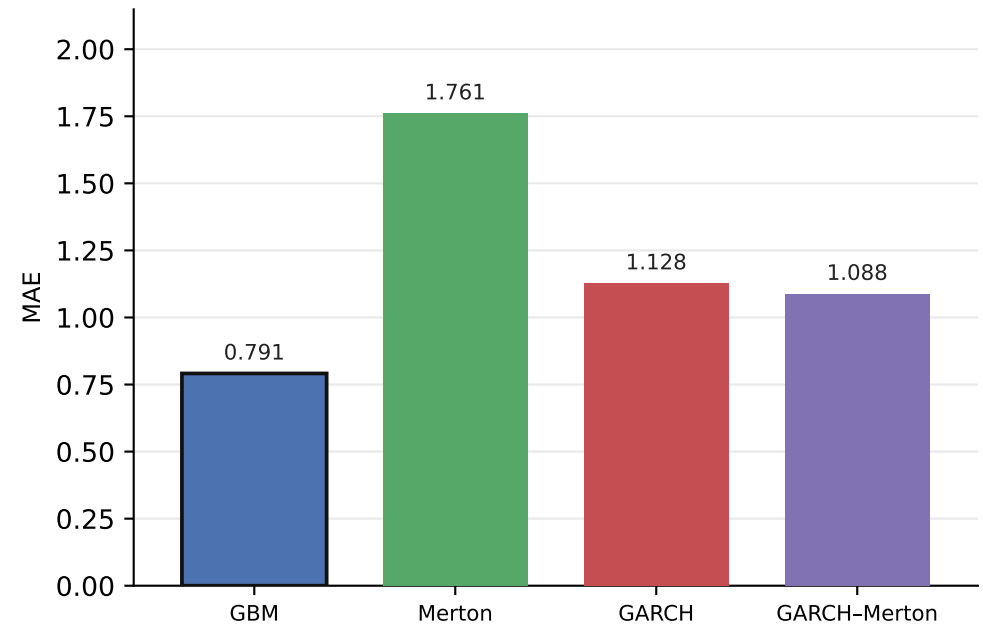
1-day · minutely rolling



5-day · hourly rolling



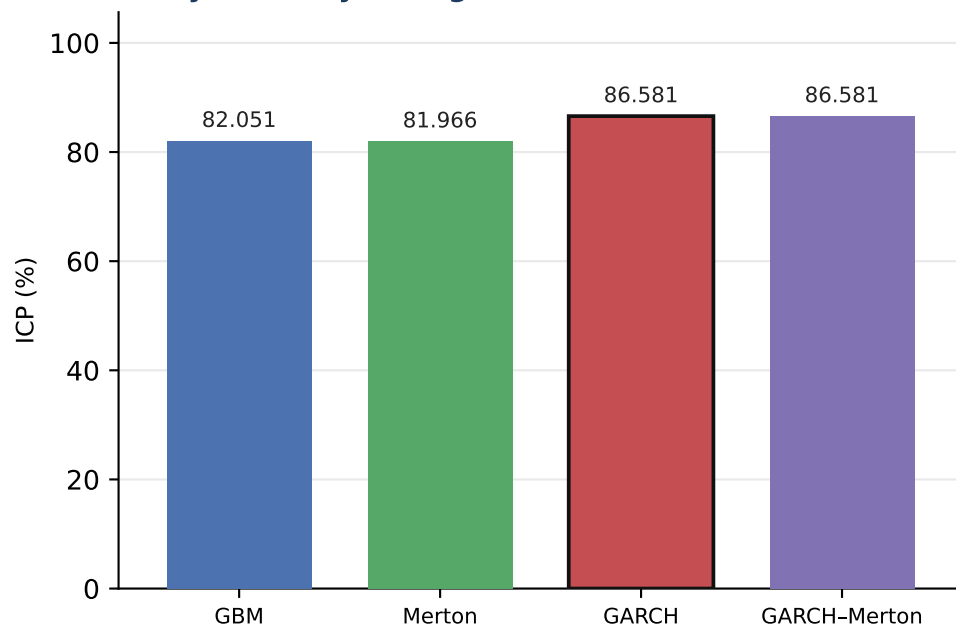
5-day · minutely rolling



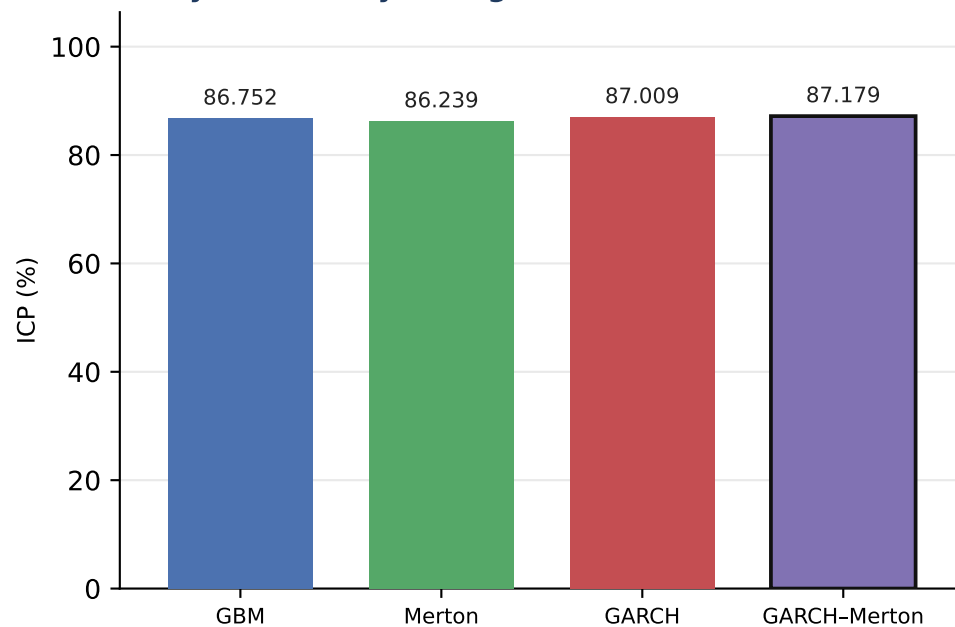
Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

ICP (25th-75th band) · mean of SPY / AAPL / MSFT

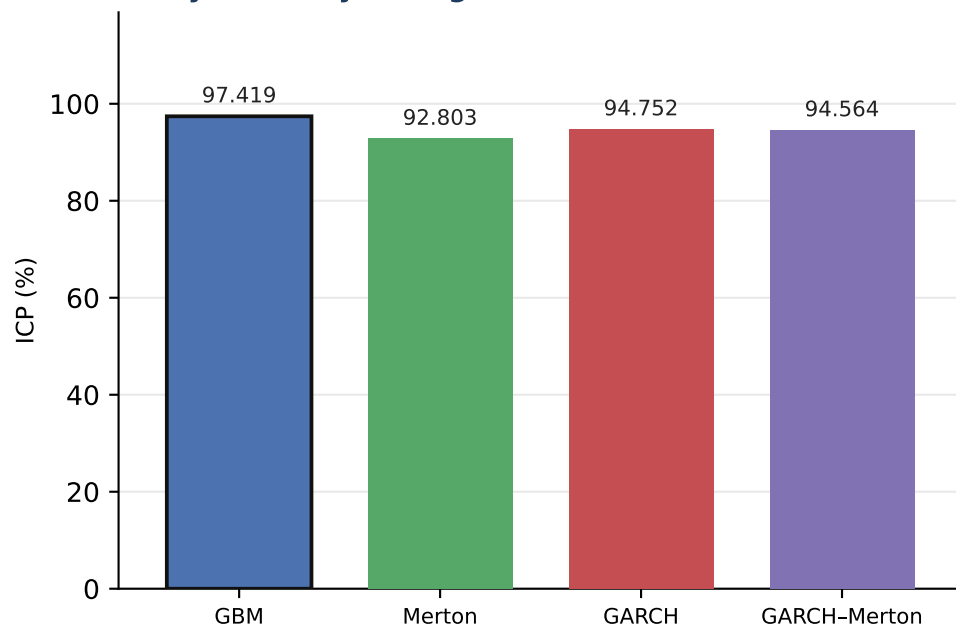
1-day · hourly rolling



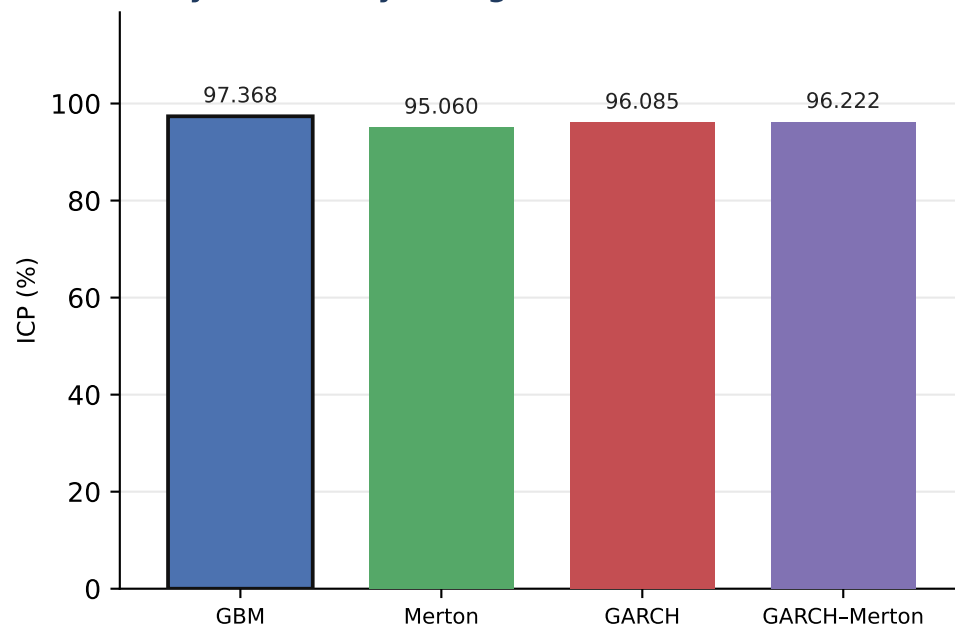
1-day · minutely rolling



5-day · hourly rolling



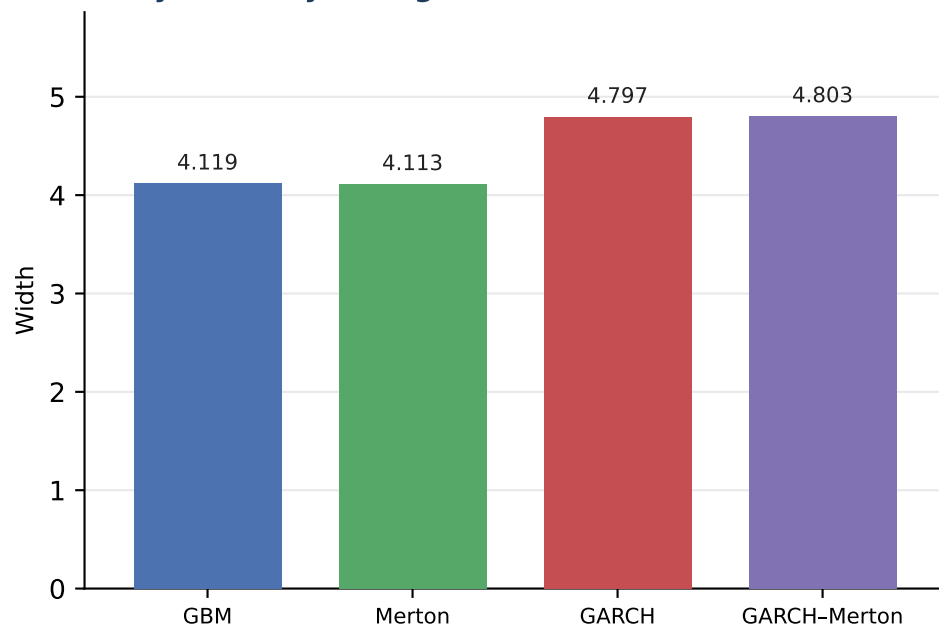
5-day · minutely rolling



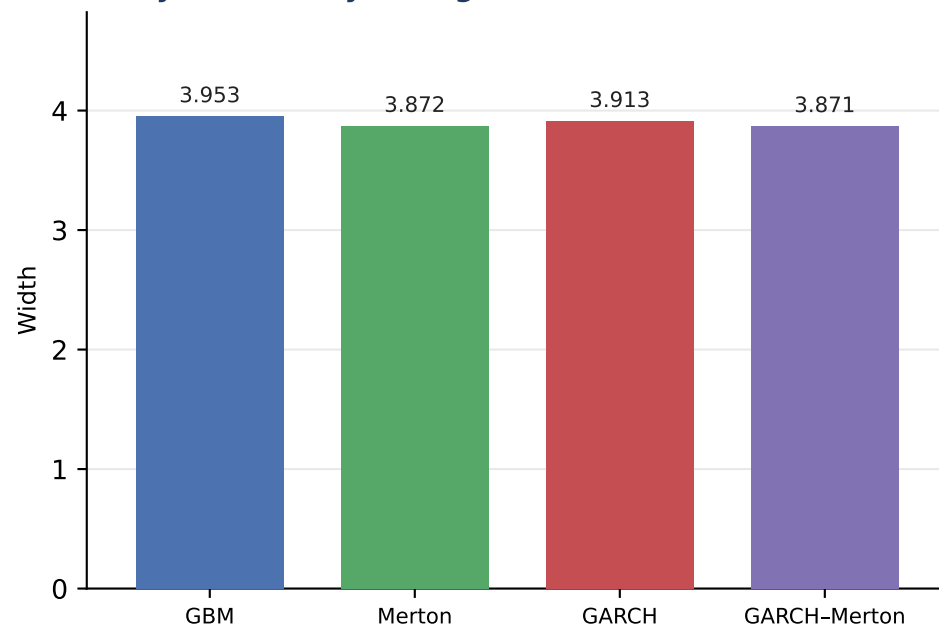
Highest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

Average 25th-75th band width · mean of SPY / AAPL / MSFT

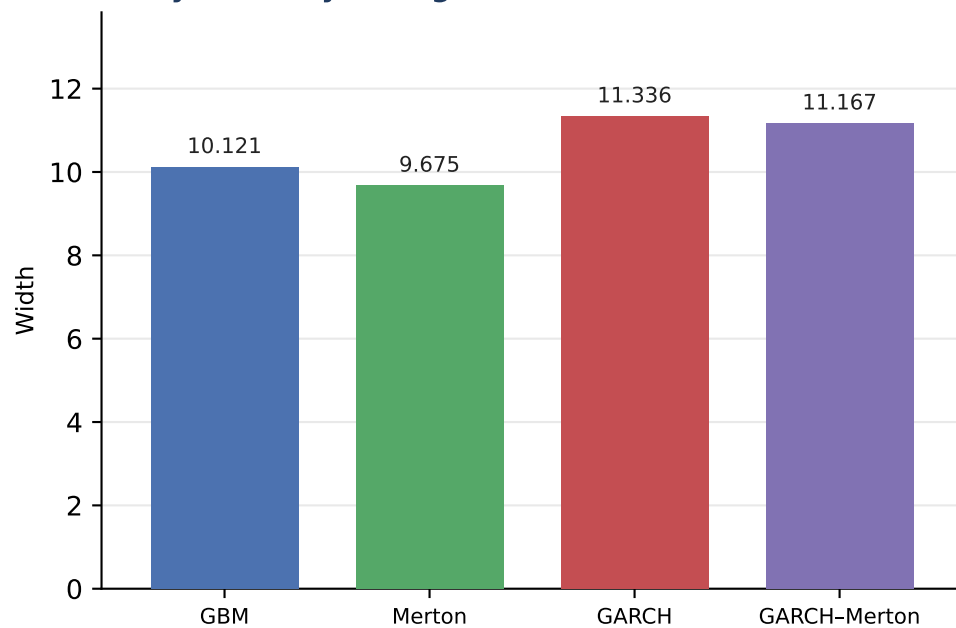
1-day · hourly rolling



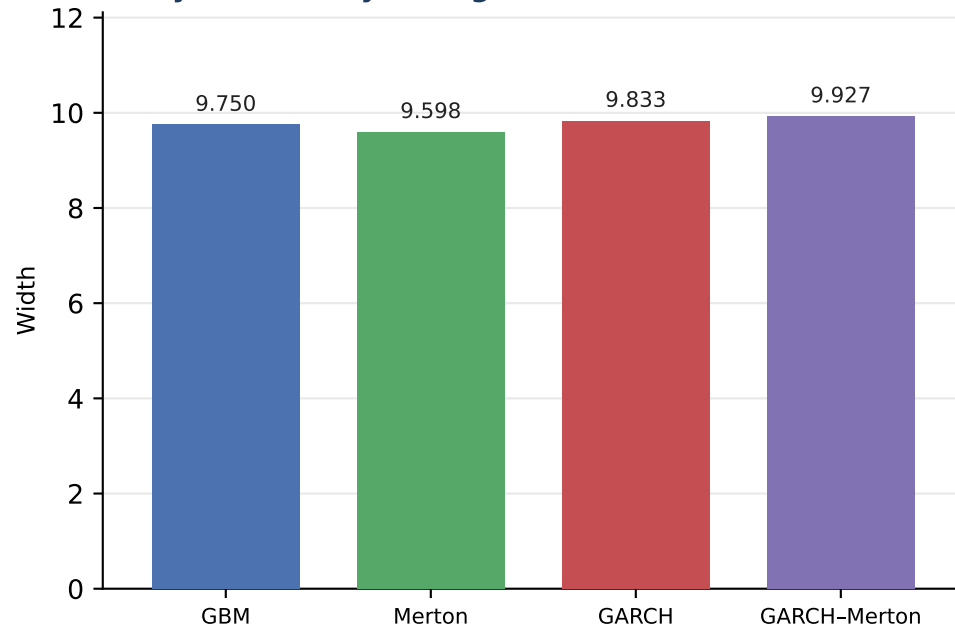
1-day · minutely rolling



5-day · hourly rolling

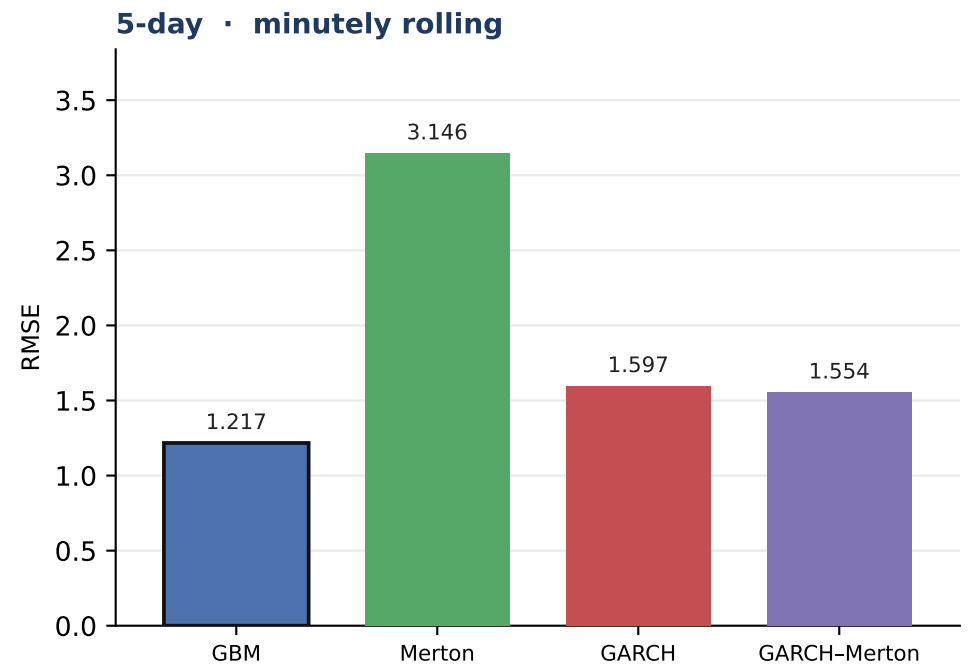
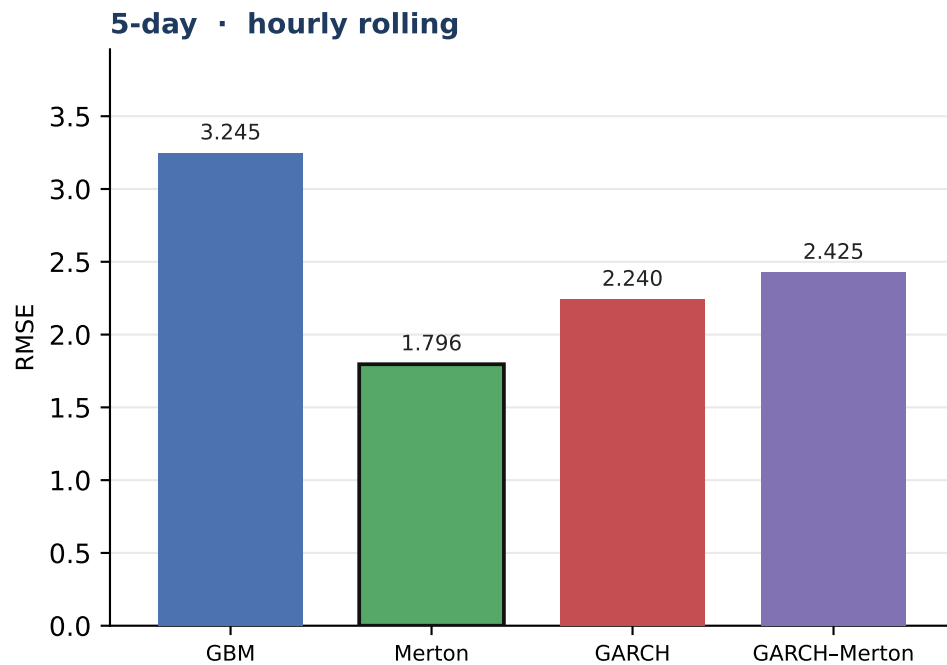
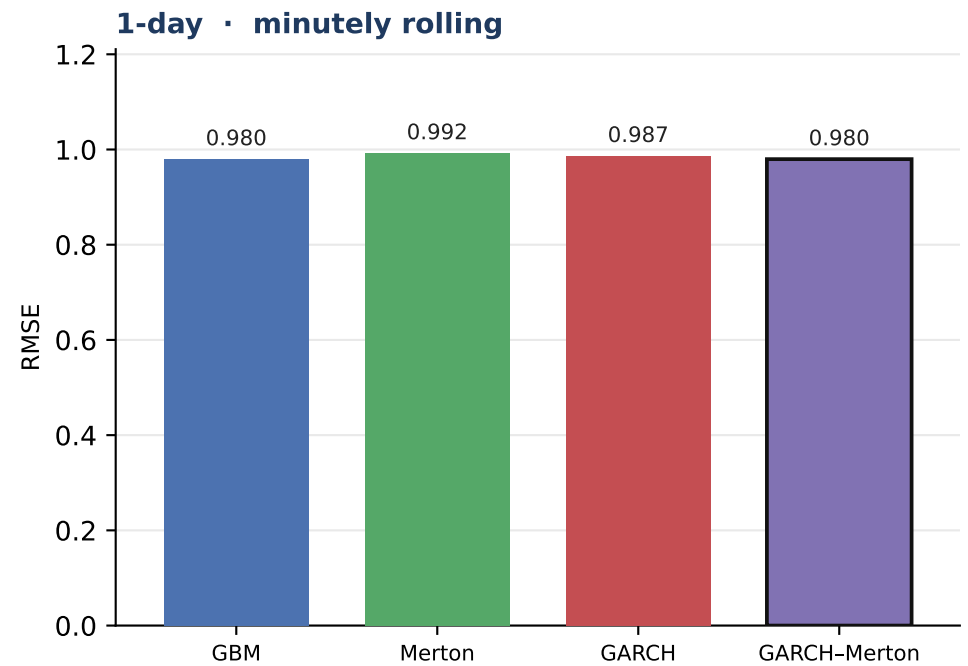
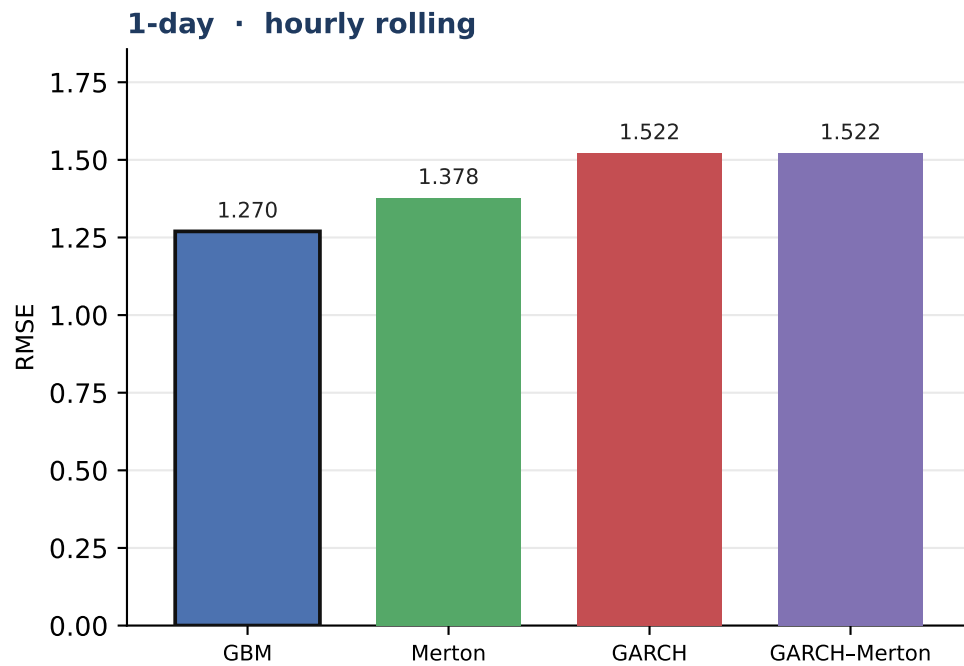


5-day · minutely rolling



Narrower is not a ranking · Heston and Heston-Merton omitted (see tables PDF).

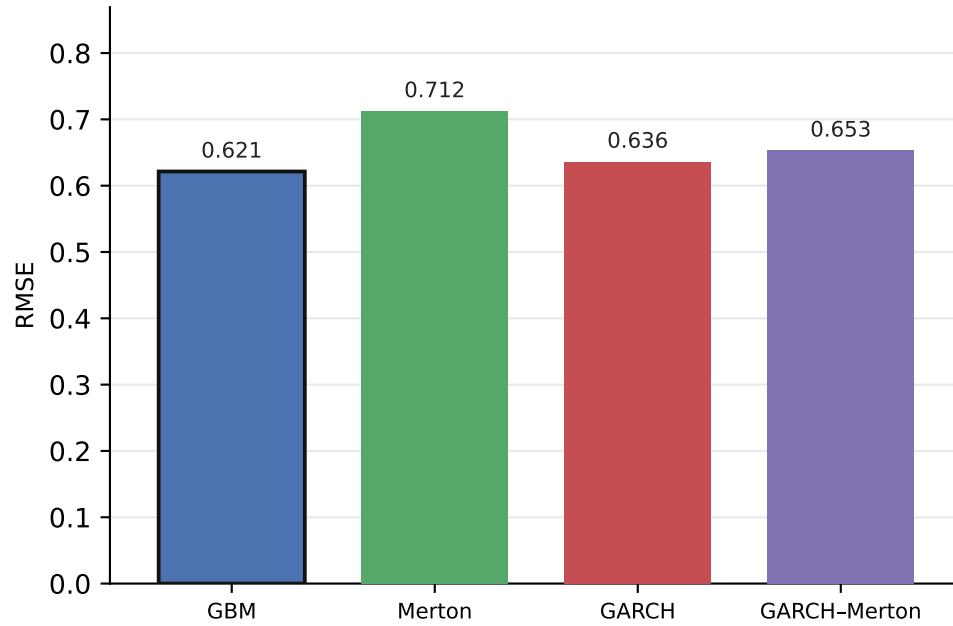
RMSE of median path vs S_t · SPY



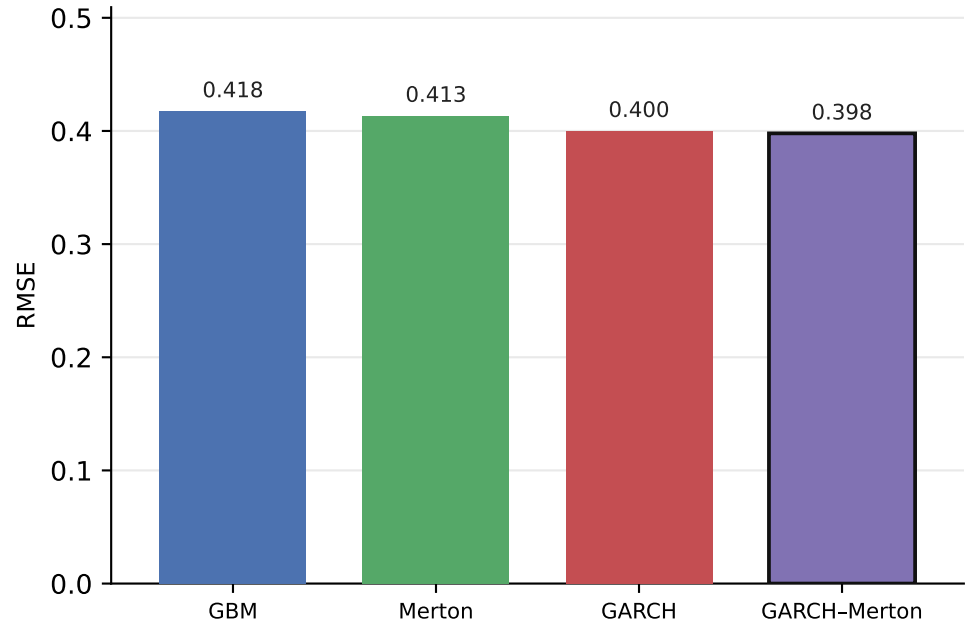
Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

RMSE of median path vs S_t · AAPL

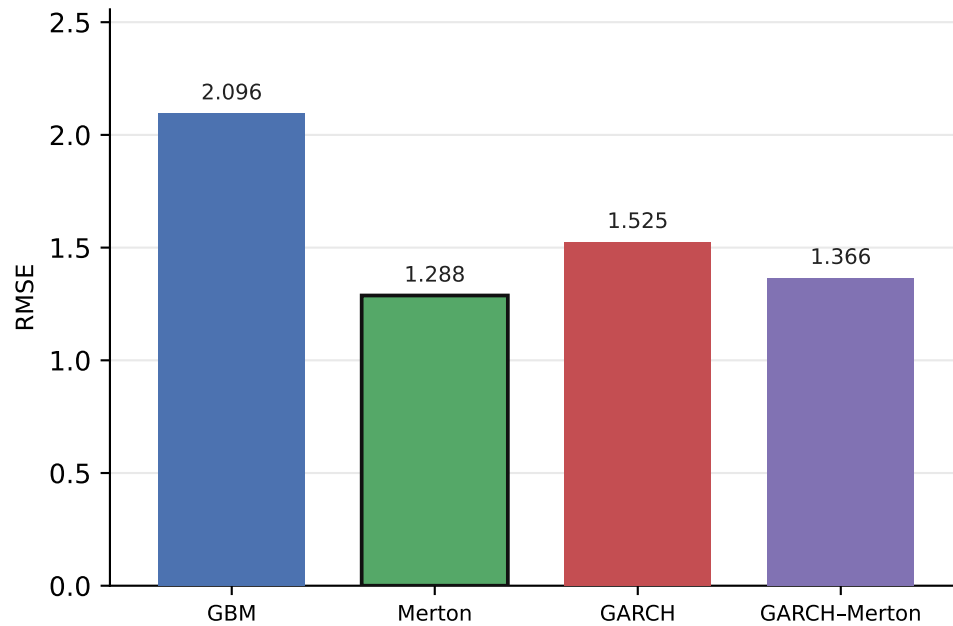
1-day · hourly rolling



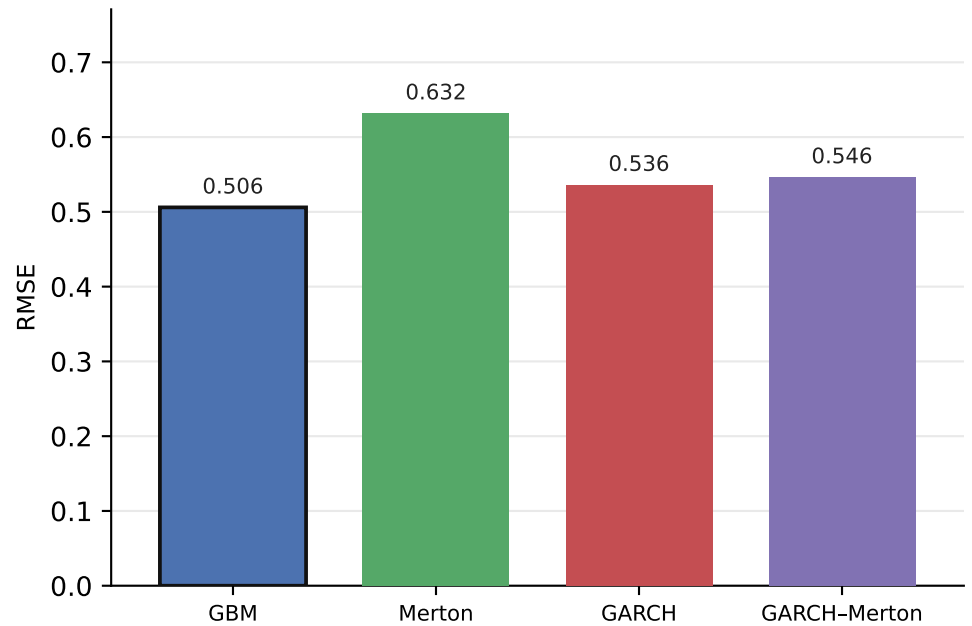
1-day · minutely rolling



5-day · hourly rolling



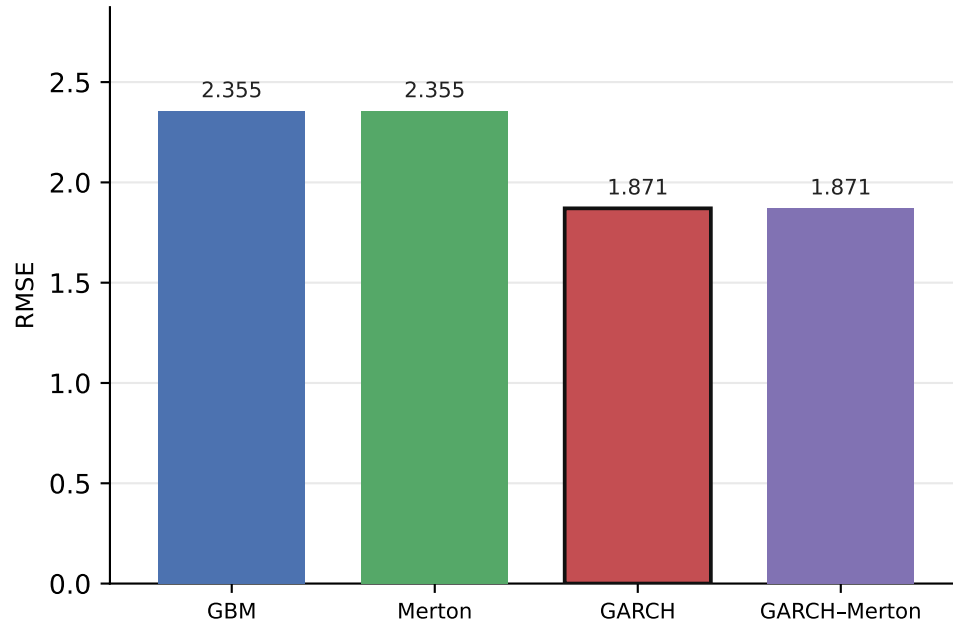
5-day · minutely rolling



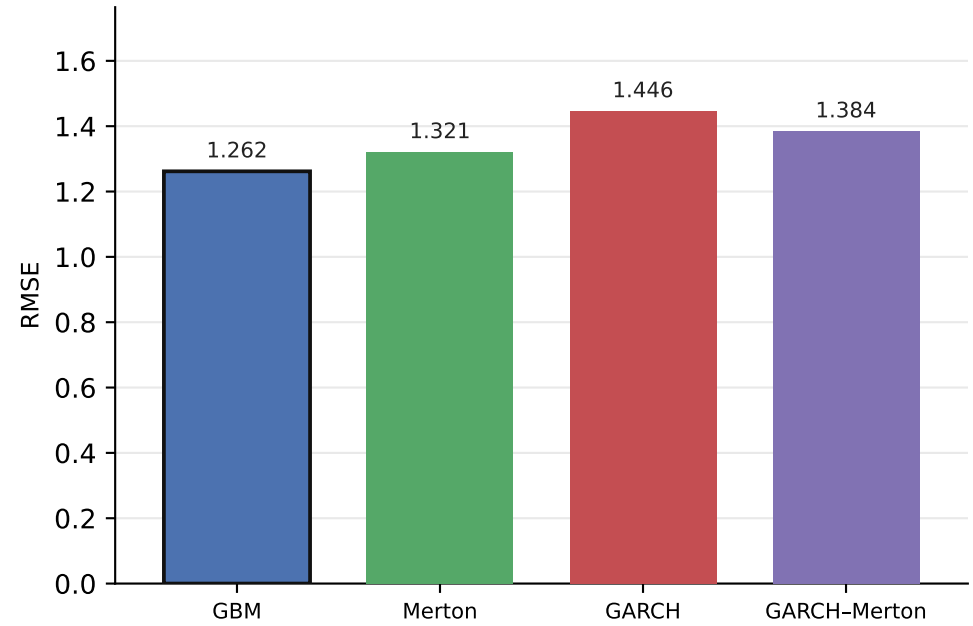
Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

RMSE of median path vs S_t · MSFT

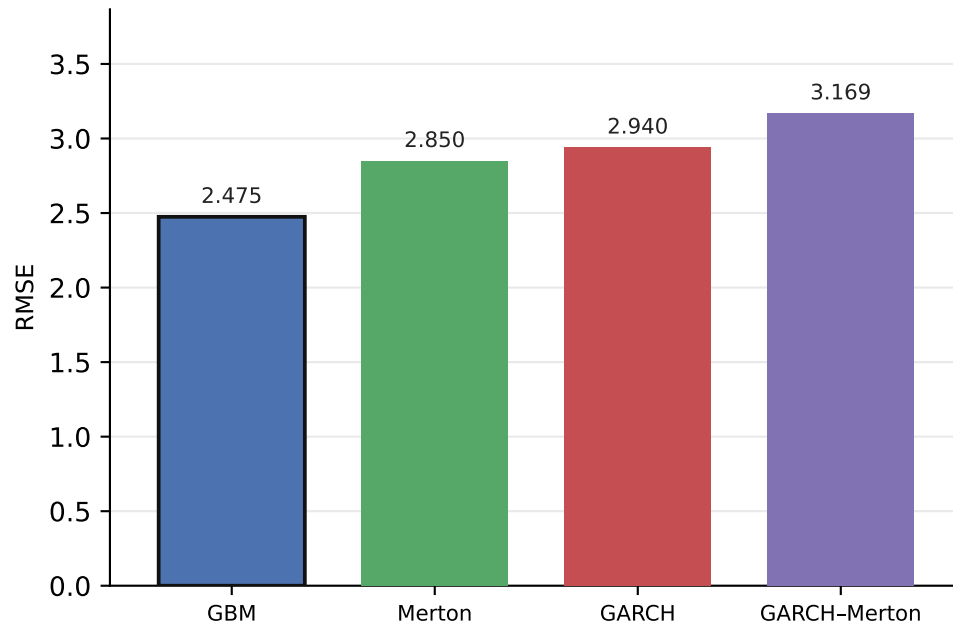
1-day · hourly rolling



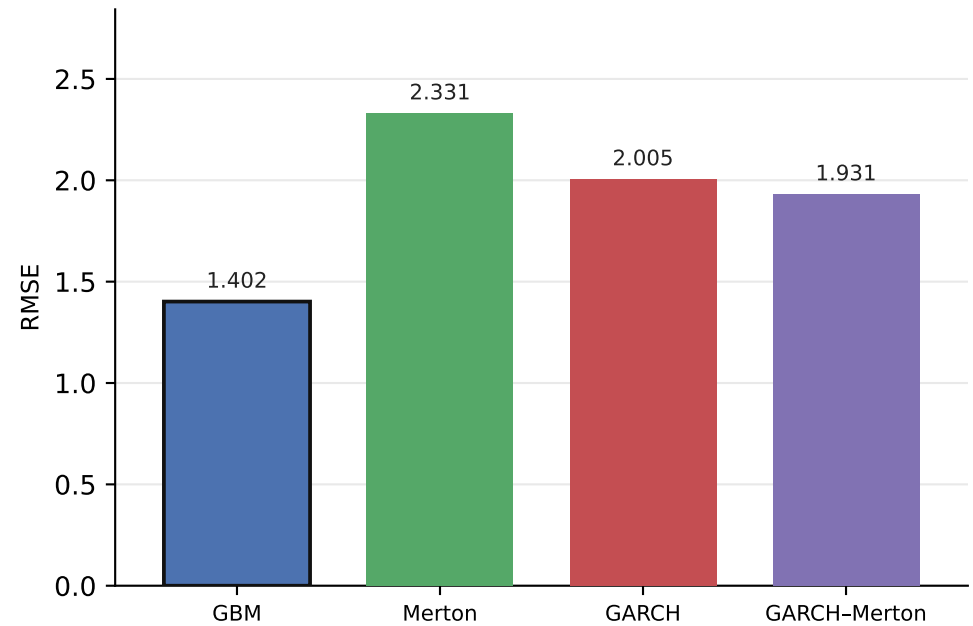
1-day · minutely rolling



5-day · hourly rolling



5-day · minutely rolling



Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).